

Key Challenges in Hybrid AI Architectures (TSM)

This document summarizes the major challenges involved in building a hybrid sequence model combining Transformer, Mamba, and Hyena architectures.

1. Representation Mismatch

Different architectures represent information differently. Transformers use attention-based relational representations, Mamba uses compressed state memory, and Hyena uses convolution-based pattern signals. Combining these heterogeneous outputs is difficult.

2. Fusion Problem

Determining how to combine outputs from multiple branches is challenging. Improper fusion can cause one component to dominate while others become ineffective. Adaptive gating is required but complex to design.

3. Training Instability

Different components learn at different speeds. This leads to gradient conflicts, unstable convergence, and difficulty in training the model effectively.

4. Computational Complexity

Running multiple architectures in parallel increases computational cost, memory usage, and engineering complexity, making scaling difficult.

5. Over-Engineering / Practical Trade-off

Transformers already perform well in many tasks. Adding multiple components may not justify the added complexity unless significant gains are achieved.

6. Interaction Problem (Core Challenge)

The biggest challenge is not building individual models, but making them work together intelligently. Designing coordination mechanisms between memory, reasoning, and pattern extraction is the core research problem.